

Priyanka Anil Lakariya

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EDUCATION

Northeastern University | *Master of Science in Robotics* | **GPA:** 3.86/4

Sep 2025 - May 2027

Savitribai Phule Pune University | *Bachelor of Engineering in Mechanical* | **GPA:** 3.5/4

Sep 2019 - Jul 2023

SKILLS

Languages & Middleware	: Python, C++, Embedded C, SQL, MATLAB, ROS/ROS2
Robotics	: RTAB, ORB-SLAM3, PCL, OpenCV, PyTorch, TensorFlow, YOLO, AprilTag, Nav2, AMCL, DWA, RViz
Software	: Gazebo, MuJoCo, SolidWorks, Simulink
Hardware	: Modbus, TCP/IP, I2C, LiDAR, RGB-D Camera, IMU/Gyroscope, Motor control, Sensor Calibration
Tools	: Linux, Git, scikit-learn, NumPy, SciPy, Pandas, Matplotlib
Certification	: OpenCV Bootcamp, Autonomous Robots (NVIDIA), Foundations of IoT Systems & Industrial Automation

EXPERIENCE

Innomotics India Pvt. Ltd. (formerly Siemens Large Drive) | *Manufacturing Professional*

Jul 2023 - Jul 2025

- Supervised end-to-end production of **3.3KV–11KV** medium-voltage motors, coordinating supplier timelines, tracking lead times, and generating production schedules to scale monthly output from **10 to 25** units
- Analyzed **2** years of **1000+** order records using **Advanced Excel** to classify A/B/C material demand, implementing **Kanban/JIT** to improve monthly output by **100%**
- Automated **MIS reporting** by centralizing cross-department updates in one dashboard, reducing report cycle time by **40%**
- Redesigned kitting trolley layout using **shadow boards**, barcode scanning, and material kitting tracker, reducing manual search time by **25%** and enabling real-time material stage visibility across assembly stations
- Tested and validated **3.3KV–11KV** VFDs with **hardware configuration**, ensuring product reliability across assembly

Wissenquelle - OneStop Ai | *Robotics Project Intern*

Feb 2022 - Apr 2022

- Derived **DH parameters** and computed **forward kinematics** for 6-DOF PUMA 560, 3-DOF planar manipulator, and 4-DOF SCARA in **RoboAnalyzer**, determining end-effector poses across multiple joint configurations
- Validated kinematic models against simulation, verifying spatial consistency of joint-space to task-space mappings

PROJECTS

GPS-Denied Drone SLAM for Underground Mine Mapping | *ROS2, LIO-SAM, RTAB-Map, PCL, Gazebo, Nav2*

[GitHub](#)

- Integrated LIO-SAM + IMU pipeline to achieve globally consistent 3D reconstruction across cave meshes
- Constructed RTAB-Map loop closure pipeline across **10,578 nodes**, generating a complete 3D point cloud reconstruction of the tunnel at **0.01 m** voxel resolution
- Deployed frontier-based exploration via Nav2 + OctoMap, achieving 100% waypoint coverage across cave topology
- Trained k-NN/SVM NIR spectral classifier identifying 5 mineral types at **81–93% confidence**, georeferencing **2,617 detection points** as 3D markers on the live OctoMap

Real-Time Pedestrian Trajectory Prediction | *RTMPose, SMPL, Social-LSTM, CVAE, PyTorch*

[GitHub](#)

- Engineered dual-stream architecture combining **Social-LSTM** with Transformer pose encoder, achieving **35–46%** error reduction on JTA dataset and **0.67m minADE** on ETH/UCY benchmark
- Deployed **RTMPose + SMPL** for 3D human pose estimation from at **30 FPS**, extracting 17 keypoints to predict pedestrian intent **0.8–1.0s earlier**
- Developed **CVAE decoder** generating K=20 multimodal trajectories over a 4.8s horizon

3D Scene Reconstruction via Structure-from-Motion | *SIFT, Grabcut, RANSAC, EPnP, Bundle Adjustment, OpenCV*

[GitHub](#)

- Developed a SfM pipeline applying SIFT with GrabCut, extracting **1199 keypoints per image** across a 62-image dataset
- Implemented **Lowe's ratio test** and RANSAC estimation to reject outliers, achieving $\geq 70\%$ inlier ratios
- Reconstructed camera trajectory via Essential Matrix decomposition, triangulating a **12,968-point** sparse point cloud
- Achieved **0.83 px mean reprojection error** by sparse bundle adjustment with COO Jacobian, producing an **8,672-point** point cloud in Meshlab

Kinematic Modeling and Trajectory Tracking for UR5 Collaborative Robot

[GitHub](#)

- Implemented forward kinematics for **6-DOF UR5 manipulator** using DH convention and developed **Newton-Raphson IK solver** with damped least-squares on 6x6 Jacobian, achieving **sub-micron** pose accuracy in 29 iterations
- Tracked circular and **infinity trajectories** across 157 waypoints maintaining fixed end-effector orientation, achieving **sub-millimeter** tracking accuracy across all waypoints
- Validated joint-space solutions against **UR5 joint limits**, with warm-start convergence in **2–5 iterations** enabling real-time trajectory execution